

Osama Khaled

📍 Cairo, Egypt

✉ osamakhaled132000@gmail.com

☎ +201115856439

🌐 [linkedin.com/in/osama-khaled-739984218](https://www.linkedin.com/in/osama-khaled-739984218)

🐙 github.com/osamakhaled123

🤖 huggingface.co/osamakhaledML9

Summary

Results-driven Machine Learning Engineer expertise in Data Science, Computer Vision, NLP, and Deep Learning, LLMs, RAG and Agentic AI. With hands-on experience in deploying ML models for mental health classification, EEG analysis, and image processing. Passionate and interesting about solving complex problems through innovative ML solutions.

Work Experience

Machine Learning Research Internship, NajahNow Corporation

Aug 2022 – Oct 2022

- Built a mental health classification model using Arabic social media data, created benchmark dataset of 8,000+ entries and fine-tuned AraBERT/MarBERT, improving classification accuracy by 15% over baseline models.
- Collaborated with cross-functional teams to integrate the model into a mental health monitoring tool, reducing manual analysis time by 30%.

Education

B.Sc. of Computers and Artificial Intelligence, Cairo University

Sept 2019 – July 2023

- **GPA:** 3.5/4.0 (Excellent) | Ranked 10th in the Department of AI
- **Relevant Courses:** Machine Learning, Deep Learning and Natural Language Processing Specializations from DeepLearning.AI, MLOps from DataCamp, Intro to NLP and Intro to LLM and Intro to AI Agent and Agentic AI from 365DataScience, Data Analysis Nanodegree from Udacity.

Projects

Neural Style Transfer Implementation → |PyTorch, torchvision, Numpy|

- An implementation of the neural style transfer technique using VGG19 model.
- Loading pre-trained VGG19 weights, calculating weighted content and style loss functions, and iteratively updating the target image to minimize the combined loss. Loss decreased about 7.5 times of the loss at the beginning of the training process.

Feature Extraction and Image Classification → |Numpy|

- Built a neural network from scratch using NumPy, achieving 86% accuracy on the MNIST dataset.
- Extracted centroid coordinates from 4x4 sub-images where each sub-image size is 7x7, and used them as features for classification.

EEG Signal Classification for Hand Movement → |TensorFlow, Numpy, Sklearn|

- Engineered a Random Forest model with 89.5% accuracy to classify hand movements from EEG signals, reducing noise using wavelet transforms.
- Visualized frequency-domain features to identify key biomarkers for motor cortex activity.

Transformer from Scratch for Summarization → |Pytorch, Hugging Face|

- Implemented a complete Transformer-based sequence-to-sequence model from scratch in PyTorch for CNN/DailyMail summarization, without relying on pretrained libraries.
- Designed an end-to-end data preprocessing pipeline including custom tokenization, text cleaning, padding/truncation, and dataset filtering to ensure high-quality training data.
- Achieved a BLEU score of 22, reducing training loss from 24.4 → 8.37 over 5 epochs, while implementing advanced techniques such as beam search decoding, attention masking, weight tying, and sequence-level inference, all from scratch, using just Pytorch.

Fine-tuning LLaMA-3.2-3B with QLoRA → |Transformers, Unsloth, TRL, PEFT, BitsandBytes|

- Implemented parameter-efficient fine-tuning of LLaMA-3.2-3B using QLoRA for 2 epochs with unsloth to speed up training, on CNN/DailyMail while maintaining semantic performance (BERTScore F1: 0.801) under 4-bit quantization.
- Optimized inference using beam search decoding to enhance generation stability and improve summary coherence.
- Built and deployed a production-ready summarization system via Gradio + Hugging Face Spaces, integrating quantized base model and LoRA adapters.

Visual Question Answering with BLIP → |PyTorch, datasets, PIL|

- Trained BLIP (Bootstrapped Language-Image Pre-training) model for Image Question Answering, allowing users to ask questions about visual content (Images) and receive context-aware responses.
- Achieved an accuracy of 87% on validation set, marking clear improvement over existing models on market.

Technologies

- **Languages:** Python, C++, SQL
- **Frameworks:** TensorFlow, PyTorch, Keras, Scikit-learn, FastAPI, LangChain, LangGraph, Pinecone
- **Libraries:** Numpy, Pandas, Hugging Face, OpenCV, PIL, Spacy, Scipy, re, NLTK
- **Tools:** PyCharm, AWS, Docker, Visual Studio Code, Tableau, DataSpell, Jupyter Notebook, Kaggle Notebook, Google Colab

Soft Skills

- Problem-solving, teamwork, communication, time management, Critical Thinking, Creativity, and Work Ethic.